



Review of imaging buffers used in stochastic optical reconstruction microscopy

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ARTICLE INFO

Article history:

Received 30 August 2024

Revised 19 November 2024

Accepted 25 November 2024

Available online 26 November 2024

Keywords:

Single-molecule localization microscopy

Stochastic optical reconstruction microscopy

Photo-switching

Photobleaching

Imaging buffer

ABSTRACT

Stochastic optical reconstruction microscopy (STORM), as a typical technique of single-molecule localization microscopy (SMLM), has overcome the diffraction limit by randomly switching fluorophores between fluorescent and dark states, allowing for the precise localization of isolated emission patterns and the super-resolution reconstruction from millions of localized positions of single fluorophores. A critical factor influencing localization precision is the photo-switching behavior of fluorophores, which is affected by the imaging buffer. The imaging buffer typically comprises oxygen scavengers, photo-switching reagents, and refractive index regulators. Oxygen scavengers help prevent photobleaching, photo-switching reagents assist in facilitating the conversion of fluorophores, and refractive index regulators are used to adjust the refractive index of the solution. The synergistic interaction of these components promotes stable blinking of fluorophores, reduces irreversible photobleaching, and thereby ensures high-quality super-resolution imaging. This review provides a comprehensive overview of the essential compositions and functionalities of imaging buffers used in STORM, serving as a valuable resource for researchers seeking to select appropriate imaging buffers for their experiments.

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1. Introduction

Single-molecule localization microscopy (SMLM), typically including photo-activated localization microscopy (PALM) [1], fluorescence PALM (fPALM) [2], stochastic optical reconstruction microscopy (STORM) [3], direct STORM (dSTORM) [4], points accumulation for imaging in nanoscale topography (PAINT) [5], DNA-PAINT [6], and minimal photon fluxes (MINFLUX) [7], has revolutionized the field of biological imaging by surpassing the diffraction limit and providing detailed insights into the nanoscale structures and organizations of biological samples [8–12]. In this technique, only a small subset of fluorophores is activated at any given time, allowing for the detection of isolated emission patterns referred to as point spread functions (PSFs). These emission patterns are precisely localized using mathematical models to determine the three-dimensional (3D) positions of the fluorophores. Subsequently, millions of these localized positions are accumulated to reconstruct a super-resolution image (Fig. 1) [13–15]. In SMLM, photo-switchable fluorophores alternate between an emissive (“on”) state and a non-emissive (“off”) state, a process known as photo-switching [16–18].

On basis of the type of the fluorophores used and the method employed to induce photo-switching, SMLM approaches can be categorized into different strategies.

PALM and its variant, fPALM, employ genetically engineered photoactivatable fluorescent proteins that can be activated using ultraviolet (UV) light, facilitating effective photo-switching [1,2,19]. In contrast, STORM utilizes synthetic fluorophores, such as cyanine dye pairs, which undergo reversible photo-switching in suitable imaging buffers [3]. dSTORM further streamlines the system by enabling the use of commercially available fluorophores that are photo-switchable under specific imaging buffer and excitation conditions [4].

PAINT achieves high-resolution imaging by utilizing freely diffusing fluorescent probes that form transient bonds with target molecules [5]. In this process, a freely diffusing fluorescent molecule attaches to the target structure, generating a concentrated burst of brightness at the binding site from the emitted photons. This temporary interaction allows for the continuous replacement of old, bleached probes with fresh, unbleached ones. Afterwards, DNA-PAINT enhances super-resolution imaging by linking short DNA strands to the targets, where complementary fluorescently labeled strands bind and unbind, creating a blinking effect [6,20–23].

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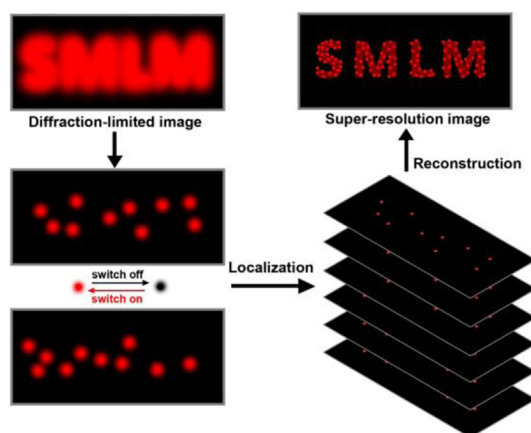


Fig. 1. Basic concept of SMLM. Reproduced with permission [18]. Copyright 2018, American Chemical Society.

Live-cell imaging requires stringent experimental conditions to ensure cell viability while facilitating effective blinking of fluorescent probes. Additionally, challenges such as photobleaching and background further complicate the imaging process. MINFLUX offers a novel approach for live-cell super-resolution imaging, where a doughnut-shaped excitation beam with a central zero-intensity point is utilized to selectively activate photoactivatable fluorescent proteins located outside the center. The method begins by scanning for fluorophores; if none are detected, high-intensity UV light activates the fluorophores. Once a fluorophore is identified, the UV light is turned off to minimize further activation and photon emission. This approach enables rapid imaging in confined regions, allowing for precise localization of fluorophores with minimal photon emission, thereby significantly reducing photobleaching and background in live-cell imaging [7,24,25].

The principles of various SMLM strategies underscore the importance of the photo-switching behavior of fluorophores as a critical factor for achieving high localization precision, which is significantly influenced by the imaging buffer [26]. Due to space limitations, this review will focus on the imaging buffer used in STORM, which typically contains oxygen scavengers, photo-switching reagents, and refractive index regulators. Oxygen scavengers help resist photobleaching, photo-switching reagents aid in facilitating the conversion of fluorophores, while refractive index regulators are utilized to adjust the refractive index of the solution. The synergistic interplay of these components sustains stable blinking of fluorophores, mitigates irreversible photobleaching, and thereby ensures high-quality super-resolution imaging. This review elucidates the key compositions and working mechanisms of typical imaging buffers used in STORM to assist researchers in selecting appropriate imaging buffers for their experiments.

2. Oxygen scavengers

Oxygen plays a pivotal role in inducing photobleaching of a fluorophore during the photo-oxidation process, signifying an irreversible transition from the emissive state to the non-emissive state [27]. This photobleaching reaction can be described by two major pathways [28].

The first pathway occurs when dissolved oxygen is in close proximity to the organic fluorophore (Fig. 2A). Upon excitation, fluorophores transition from the ground state (1S_0) to the excited singlet state (1S_1), with a portion subsequently converting to the triplet state (3T_1) through intersystem crossing (ISC). At this point, molecular oxygen (O_2) can be converted into reactive singlet oxygen (1O_2) [29–31]. The direct radiative transition from the S_1 state

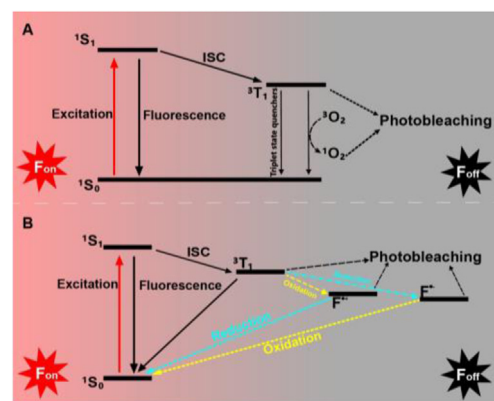


Fig. 2. Photo-switching process of common organic fluorophores. (A) In the “on” state, fluorophores transition from 1S_0 to 1S_1 . From the 1S_1 state, fluorophores can return to the 1S_0 state while emitting fluorescence through radiative processes, or they may undergo nonradiative vibrational relaxation to the 1S_0 state, a process also referred to as thermal deactivation. Alternatively, they can experience ISC to a non-fluorescent state 3T_1 . The 3T_1 state can be quenched to the 1S_0 state by triplet-triplet energy transfer with 3O_2 to generate 1O_2 , resulting in photobleaching. (B) Redox-active buffers can be used to depopulate the 3T_1 state by electron transfer either through oxidation forming a radical cation (F^+) or through reduction yielding a radical anion (F^-). The fluorophores then return to the 1S_0 state through inverse redox reactions.

to the S_0 state results in the emission of a photon with lower energy and longer wavelength, known as fluorescence. Singlet oxygen can directly oxidize the fluorophore, giving rise to photobleaching, or it can facilitate the formation of reactive oxygen species (ROS) such as peroxy radicals ($RO_2^{\cdot}$), superoxide anions ($O_2^{\cdot-}$), and hydroxyl radicals ($OH^{\cdot}$), which further promote photobleaching through free radical reactions. In addition to promoting photobleaching, ROS can damage biological samples by oxidizing cellular components, compromising the structural integrity during imaging [32]. In this pathway, singlet oxygen acts as the primary agent triggering photobleaching, while the triplet state of the fluorophores serves as a precursor to this reaction. Therefore, removing dissolved oxygen from the vicinity of the fluorophore can effectively slow down photobleaching.

The second pathway occurs through the formation of reactive short-lived radicals ($F^{\cdot}$) generated by excited-state-mediated photoionization (Fig. 2B). The higher excited singlet state (1S_n) and triplet state (3T_n) are closely associated with the photoionization rate. Transitions to the triplet states (3T_1 and 3T_n) promote electron transfer reactions, resulting in photobleaching and irreversible degradation of fluorophores due to radical formation. The rapid return to the singlet ground state is essential for effectively inhibiting side reactions that contribute to photobleaching. Consequently, various combinations of reducing agents, antioxidants, and triplet state quenchers have been investigated to minimize the generation of free radicals and prevent oxygen-independent photobleaching.

To enhance the photostability of fluorophores, researchers initially utilized an enzyme-based oxygen scavenging system composed of glucose oxidase and catalase (GLOX, also known as GOD-CAT), which converts glucose and oxygen into gluconic acid and water through a two-step process (Fig. 3A) [33–40]. The GLOX system is effective at eliminating oxygen and has been widely applied in STORM [41–50]. However, inadequate coordination between the activities of glucose oxidase and catalase can lead to the accumulation of hydrogen peroxide. The transient presence of peroxide competes with sample proteins, potentially amplifying its detrimental effects on fluorophores and biomolecules [51–53].

To overcome these limitations, in 2008, Aitken *et al.* [54] introduced an oxygen scavenging system for use in STORM, comprising protocatechuic acid (PCA) and protocatechuate-3,4-dioxygenase

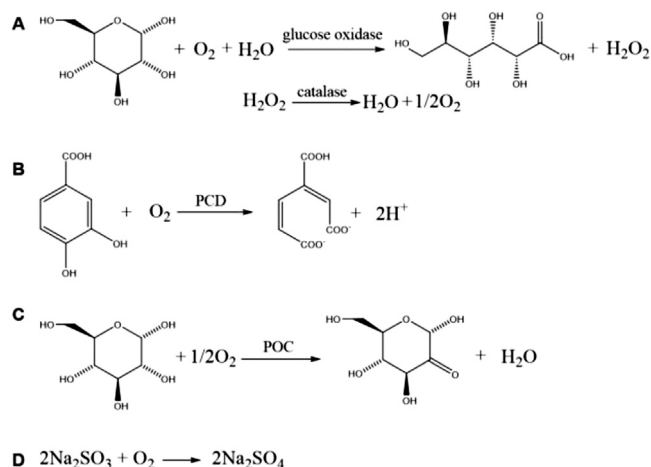


Fig. 3. Reactions in the oxygen scavenging systems. (A) Reactions in the GLOX system. (B) Reactions in the PCA+PCD system. (C) Reactions in the POC system. (D) Reactions in the system with sodium sulfite.

(PCD). The researchers quantified the initial reaction rate and steady-state O_2 concentration of the GLOX system with a simple kinetic assay of enzyme activity, and then standardized the O_2 -scavenging capability of the PCA+PCD system against the GLOX system through a dissolved oxygen assay. Experimental results illustrated that the PCA+PCD system could successfully substitute the GLOX system for oxygen scavenging (Fig. 3B). Subsequently, the researchers assessed the photophysical stability of the dyes, Cy3, Cy5, and Alexa Fluor (AF)488, in both the GLOX system and the PCA+PCD system which was supplemented with 1 mmol/L of 6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid (Trolox). In the GLOX system, Cy3 and Cy5 exhibited longer initial lifetimes with infrequent blinking and a signal-to-noise ratio (SNR) of 4.0, while AF488 demonstrated a shorter lifetime with more frequent blinking and an SNR of 2.5. In contrast, the initial lifetimes of all three dyes increased by up to 140% in the PCA+PCD system, without affecting the blinking frequency or SNR. This PCA+PCD system has since been widely adopted in STORM [12,55–59]. However, both the GLOX and PCA+PCD systems produce by-products, gluconic acid and carboxylic acid, leading to a gradual decrease in pH over time. This pH drop can trigger undesirable blinking behaviors in the fluorophores and pose challenges for long-term imaging.

To stabilize pH values, in 2012, Swoboda *et al.* [60] proposed an oxygen scavenging system that utilizes pyranose oxidase, D-glucose, and catalase (POC). This POC system produces a ketone (2-dehydro-D-glucose) as a reaction product, which does not affect the pH value during the reaction (Fig. 3C). The researchers conducted a comparative study on the photobleaching performance of the POC, GLOX, and PCA+PCD systems at starting pH values of 7.0, 7.5, and 8.0 using the fluorophores Cy3 and Cy5 in a custom-built setup for single-molecule fluorescence resonance energy transfer (FRET). Results indicated that the oxygen removal efficiencies of the GLOX and PCA+PCD systems were influenced by the initial pH value and ion concentration. In contrast, the POC system exhibited effective oxygen removal capabilities while maintaining pH stability, regardless of the initial pH value or ion concentration (Fig. 4A). Later in 2016, Nahidiazar *et al.* [61] presented the OxEA system, which incorporates the oxyrase and DL-lactate added as a substrate. Oxyrase, a sterile solution containing membrane fractions of *Escherichia coli*, is capable of removing oxygen from solutions [62]. Nahidiazar *et al.* showed that the pH value of the buffer could remain stable for at least 4 h when using the OxEA system in exposed environments (Fig. 4B), enabling fluorophores to exhibit con-

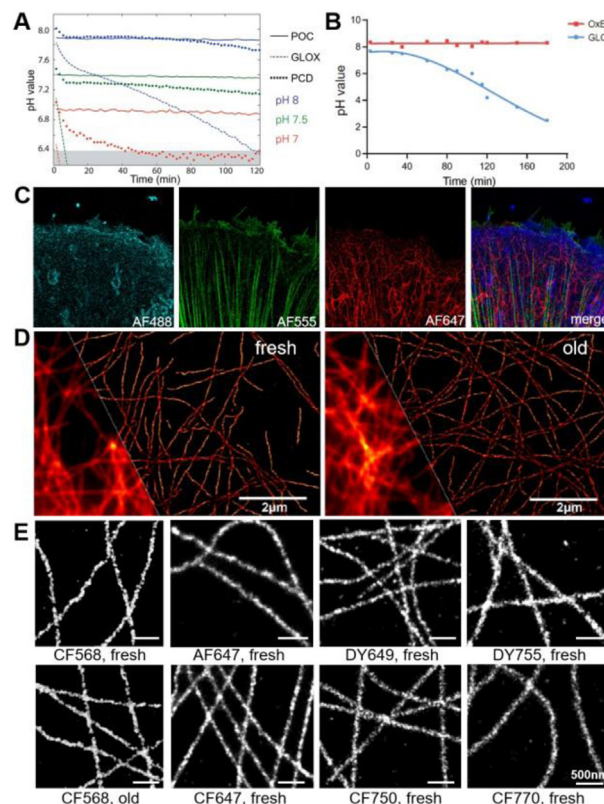


Fig. 4. Effects of oxygen scavenging system. (A) pH values across time in the POC (solid), GLOX (dashed), and PCD (dotted) systems with initial pH values of 8 (blue), 7.5 (green), and 7 (red). Reproduced with permission [60]. Copyright 2012, American Chemical Society. (B) pH values across time in the OxEA and GLOX systems. (C) Three-color super-resolution image of $\beta 4$ integrin (red), keratin (green), and plectin (blue) using the OxEA system. Reproduced with permission [61]. Copyright 2016, Public Library of Science. (D) Super-resolution images of microtubules immunostained with AF647 using fresh (left) and 28-day-old (right) imaging buffers containing sodium sulfite. The bottom left insets are diffraction-limited images. Reproduced with permission [66]. Copyright 2018, bioRxiv. (E) Super-resolution images of microtubules immunostained with different fluorophores using imaging buffers containing sodium sulfite. Here “fresh” means the buffer was prepared freshly, while “old” means the buffer was frozen for 1 week, thawed overnight in the fridge, and then used after another 4 days. Reproduced with permission [67]. Copyright 2022, American Chemical Society.

sistent blinking behaviors. Utilizing the OxEA system, Nahidiazar *et al.* achieved multi-color imaging with various fluorophores, including AF488, AF555, and AF647 (Fig. 4C). Afterwards, the OxEA system has been successfully applied in STORM [63–65].

In addition to above-mentioned enzyme-based oxygen scavenging systems, non-enzyme components can be also used to consume oxygen. In 2018, Hartwich *et al.* [66] proposed a high-refractive-index imaging buffer composed of 80%–90% glycerol, β -mercaptoethylamine (MEA), and sodium sulfite, which reacts with dissolved oxygen to generate sodium sulfate without causing acidification (Fig. 3D). This imaging buffer is cost-effective and does not require fresh preparation, allowing for super-resolution imaging even 28 days after preparation (Fig. 4D). The authors demonstrated that sodium sulfite could be also used in aqueous buffers, achieving a similar reduction in background and improvement in localization precision. In 2022, Abdelsayed *et al.* [67] utilized sodium sulfite as an oxygen scavenger in an aqueous imaging buffer, successfully capturing high-quality super-resolution images of microtubules that were immunostained separately with various fluorophores including CF568, AF647, CF647, Dylight (DY)649, CF750, DY755, and CF770 (Fig. 4E). This work highlighted the compatibility and effectiveness of the proposed imaging buffer in STORM.

3. Photo-switching reagents

Understanding the photo-switching mechanisms of fluorophores is essential for optimizing imaging buffers. In STORM imaging, cyanine and rhodamine dyes are frequently utilized due to their excellent photo-switching properties. The photo-switching mechanism of cyanine dyes involves the addition of a thiol to the polymethine bridge, which disrupts the conjugated π -electron system and transitions the fluorophore to a dark state (Fig. 5A) [68–70]. In the case of rhodamines, the excited dye is reduced by thiols in a deoxygenated buffer, resulting in a long-lived radical dark state that can be reverted to the emissive state by exposure to UV light or an oxidant (Fig. 5B) [71,72].

As mentioned above, after the removal of oxygen with an oxygen scavenging system, photo-switching reagents, such as triplet state quenchers, reducing agents, and antioxidants, are required to recover the fluorescent state [73]. In addition, photo-switching reagents help achieve high contrast between the “on” and “off” states of fluorophores and minimize the time fluorophores spend in the “on” state, thereby enhancing SNRs and the overall image quality [17,28].

In early research, thiols were commonly used as triplet-state quenchers to conduct the photo-switching mechanisms of cyanine dyes. In 2005, Bates *et al.* [74] and Heilemann *et al.* [75] independently examined the photo-switching behavior of Cy5 using imaging buffers containing β -mercaptoethanol (BME) and MEA. By 2009, Dempsey *et al.* [76] expanded this investigation to include the photo-switching mechanisms of various cyanine dyes, such as Cy5, Cy5-diethyl, Cy5.5, Cy7, and AF647, using imaging buffers that contained thiols. These thiols, including BME and MEA, facilitate the transition of cyanine dyes to a dark state, with the transition rate depending on the concentration of deprotonated thiol in solution. The conversion to the dark state is influenced by both pH value and thiol concentration, with the formation of a cyanine-thiol adduct as the dark-state product. This underscores the crucial role thiols play in the photo-switching process of these dyes. In 2011, Dempsey *et al.* [77] quantitatively characterized the switching behaviors of 26 fluorescent dyes under different buffer conditions, including a “no GLOX and thiol” condition with only phosphate-buffered saline (PBS), a “thiol only” condition with Tris-NaCl (TN) buffer containing 10 mmol/L MEA, a “GLOX only” condition with TN buffer containing GLOX, and a “GLOX and thiol” condition incorporating all the aforementioned components. The study revealed that most dyes were rapidly photobleached under the “no GLOX and thiol” condition, while optimal imaging results were achieved with all dyes in the “GLOX and thiol” buffer. This buffer, recognized for its outstanding imaging performance, has gained widespread use in STORM [78–93].

In addition to thiols, researchers have identified several other reagents that can function as triplet-state quenchers. In 2006, Rasnik *et al.* [94] compared two compounds, BME and Trolox, in FRET

imaging, and proved that Trolox effectively quenched the triplet state of Cy5, increasing its photostability by 7–12 times compared to BME. The following year, Widengren *et al.* [95] explored strategies to mitigate photobleaching using fluorescence correlation spectroscopy (FCS). The authors discovered that *N*-propyl galate (NPG) and ascorbic acid (AA) act as antioxidants, reducing photobleaching by restoring photo-ionized fluorophore to their fluorescent active state. In 2008, Vogelsang *et al.* [73] introduced a reducing and oxidizing system (ROXS) designed to enhance photostability by regenerating reaction intermediates. This system involves the addition of reducing and oxidizing agents, such as ascorbic acid and methylviologen (MV), to the buffer in micromolar to millimolar concentrations. The authors demonstrated that the ROXS system considerably improved the photostability of various fluorophores, including Cy5, AF647, Cy3B, and ATTO565.

In 2013, Vaughan *et al.* [96] validated that the fluorescence of Cy5 and its structural analogs could be reversibly quenched by tris(2-carboxyethyl)phosphine (TCEP), a commonly used reducing agent for disulfide bonds in proteins and small molecules. Using Cy5 as an example, they observed that TCEP reacts with the polymethine bridge of Cy5 through a 1,4-addition process, forming a covalent adduct that quenches its fluorescence. Upon UV illumination, the adduct dissociates, restoring Cy5 to its fluorescent state. The authors successfully obtained super-resolution images of mitochondria immunostained with AF647 using TCEP in the imaging buffer (Fig. 6A). Thereafter, TCEP-induced photo-reversible quenching has become pivotal for acquiring top-grade super-resolution images and facilitating easy-to-implement cellular internalization assays [97]. The same year, Olivier *et al.* [98] suggested the use of the polyunsaturated hydrocarbon cyclooctatetraene (COT), a triplet state quencher, to optimize the imaging performance in STORM. By imaging microtubules immunostained with AF647 in the presence of varying concentrations of COT in the imaging buffer, the authors discovered that COT can not only prevent oxygen-mediated photobleaching but also enhance the mean photon efficiency of AF647 by up to 3.5-fold in a dose-dependent manner (Fig. 6B), leading to improved localization precision. This enhancement was observed to be consistent across different types of thiols used to induce blinking, such as MEA, BME, or their mixtures. In 2014, Carlini *et al.* [99] utilized sodium borohydride (NaBH_4) to quench cell-permeable rhodamine dyes. NaBH_4 is a mild, inexpensive reducing agent widely used in chemistry, both in labs and industry [100,101]. The experimental results illustrated that the fluorescence of the dyes was quenched after reduction by NaBH_4 and subsequently recovered upon oxidation. Notably, when these fluorophores were targeted to proteins within live cells, the fluorescence recovery occurred spontaneously, without the need for exogenous chemicals or light stimulation. This process generates a greater number of photons (Fig. 6C), markedly refining the localization precision.

In 2022, Abdelsayed *et al.* [67] introduced the use of DL-dithiothreitol (DTT) as the primary reducing agent, complemented by 1,4-diazabicyclo[2.2.2]octane (DABCO) as an additional triplet-state quencher in STORM. DTT is available as a 1 mol/L solution, eliminating the need for fresh preparation from powder, as is required for MEA, and is less odorous and toxic compared to BME. DABCO is able to elevate the blinking behavior of Cy3 in Vectashield and does not affect the blinking of AF647 when added to conventional imaging buffers. In 2024, Go *et al.* [70] investigated the photo-switching capabilities of AF647 with various photo-switching reagents and proposed a unified photo-switching mechanism for fluorophores. This study revealed that photo-switching reagents characterized by weak binding energy to fluorophores and good nucleophilicity could prompt rapid photo-switching of the fluorophores, thereby enabling super-resolution imaging. On basis of this finding, the authors suggested that sodium azide (NaN_3)

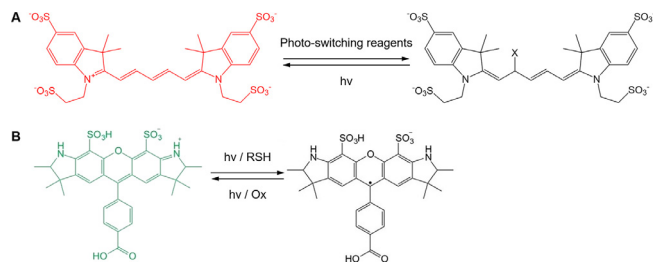


Fig. 5. Photo-switching mechanism of dyes. (A) Photo-switching reactions of cyanine fluorophores (e.g., AF647). Reproduced with permission [70]. Copyright 2024, Wiley-VCH. (B) Photo-switching reactions of rhodamine fluorophores (e.g., AF532). Reproduced with permission [72]. Copyright 2022, Wiley-VCH.

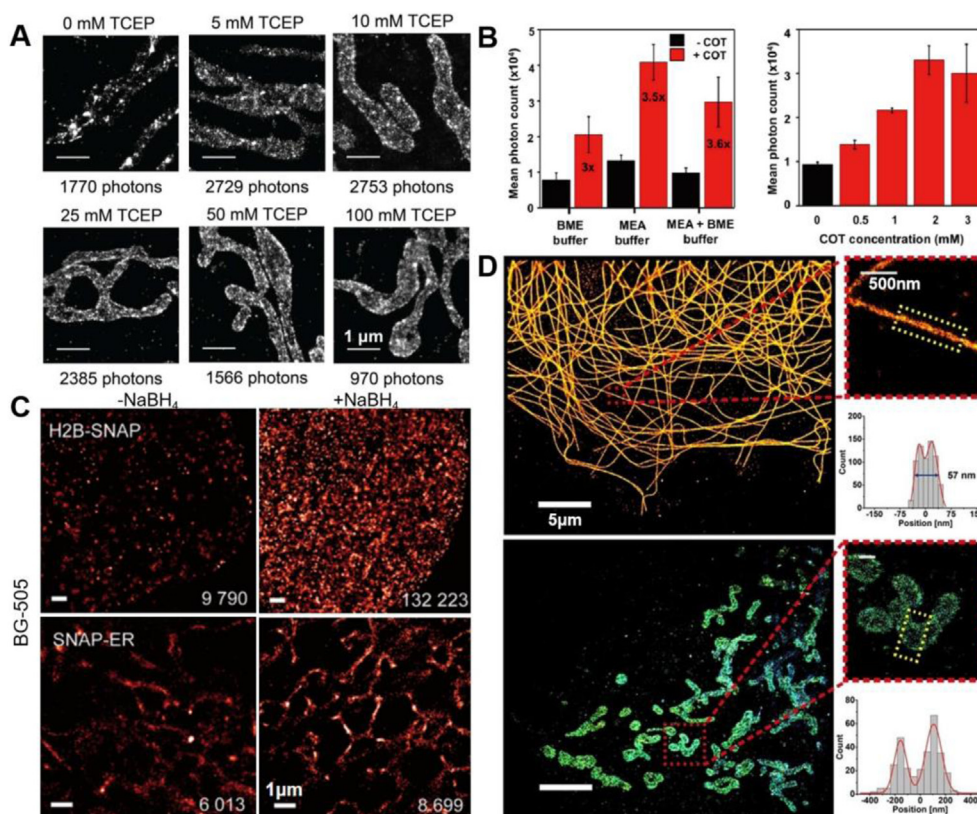


Fig. 6. Effects of photo-switching reagents. (A) Super-resolution images of mitochondria using TCEP with different concentrations. Reproduced with permission [96]. Copyright 2013, American Chemical Society. (B) Increasing photon counts in STORM using COT. Reproduced with permission [98]. Copyright 2013, Public Library of Science. (C) Super-resolution images of cells expressing the histone H2B protein fused to the SNAP tag with BG-505, with or without adding 10–50 mmol/L NaBH₄ in the imaging buffer. Numbers of localizations are shown in bottom right. Reproduced with permission [99]. Copyright 2014, Wiley-VCH. (D) Super-resolution images of microtubules (upper) and mitochondria (lower) immunostained with AF647, using 50 mmol/L NaN₃ in the imaging buffer. Reproduced with permission [70]. Copyright 2024, Wiley-VCH.

could serve as an effective photo-switching reagent, yielding high-quality super-resolution images of microtubules and mitochondria (Fig. 6D).

4. Refractive index regulators

Refractive index is another important factor to consider when preparing an imaging buffer. In STORM, high numerical aperture (NA) oil-immersion objectives are usually employed to ensure high resolution [102–105]. These immersion media, such as silicone oil (refractive index of 1.406) and oil (refractive index of 1.515), create an index mismatch with the aqueous imaging buffer which generally has a refractive index of around 1.33. This mismatch can cause spherical aberrations that significantly diminish axial resolution, especially when imaging deep into the sample [106]. Consequently, aqueous solutions are not suitable when imaging thick samples such as cell spheroids, tissues, and small animals. Matching the refractive indices of the immersion media and the imaging buffer helps reduce aberrations and suppress light scattering, thereby improving the overall imaging quality [107].

In 2007, Staudt *et al.* [108] introduced 2,2'-thiodiethanol (TDE) as a medium for imaging. TDE, a derivative of ethylene glycol, is non-toxic and can be mixed with water in any proportion, allowing for precise adjustments to the sample's refractive index. The incorporation of TDE effectively mitigates the issue of index mismatch, enabling high-resolution imaging within samples (Fig. 7A). Furthermore, the TDE solution is compatible with various cell preparation and staining techniques, acts as an antioxidant, and helps maintain the photon efficiency of most fluorophores. Therefore, numerous studies in STORM utilize TDE to optimize the refractive index of imaging buffer solutions [104,109,110].

In 2008, Huang *et al.* [111] increased the refractive index of the imaging buffer to 1.45 by incorporating either 80% glycerol and 5% glucose, or 60% sucrose and 5% glucose, enabling 3D STORM imaging of mitochondria immunostained with AF405-Cy5 with reduced spherical aberrations (Fig. 7B).

In 2013, Olivier *et al.* [112] employed Vectashield as the imaging buffer for AF647, a glycerol-based medium with a refractive index of approximately 1.45, for volumetric super-resolution imaging (Fig. 7C). Vectashield does not harden or acidify like imaging buffers that utilize enzyme-based oxygen scavenging systems, allowing for repeated imaging of the same sample without compromising the quality of the specimen. However, in 2020, Arsić *et al.* [113] observed a significant photon loss in both AF647 and AF Plus 647 (AF(+)-647) when using Vectashield, compared to the water-based buffer. Control experiments confirmed that the quenching was not due to antibody washout, but rather induced by Vectashield, which quenched fluorescence in the “on” state. The intensity of AF647 decreased to 15% of its initial value, while that of AF(+)-647 dropped to just 5%, indicating the necessity for a different buffer to achieve optimal STORM imaging.

In 2023, Lee *et al.* [114] introduced 3-pyridinemethanol (3-PM) as an imaging buffer to match the refractive index of standard immersion oil in STORM. With a refractive index of 1.545, 3-PM demonstrates good miscibility with water, allowing for precise adjustments to the sample's refractive index. It also possesses near-neutral acidity and can be mixed with HCl and NaOH without precipitation, facilitating accurate pH adjustments. The authors showed that the imaging buffer containing 3-PM delivers excellent photo-switching performance for AF647, yielding super-resolution images comparable to those obtained with conventional imaging buffers. Notably, samples prepared with the 3-PM imag-

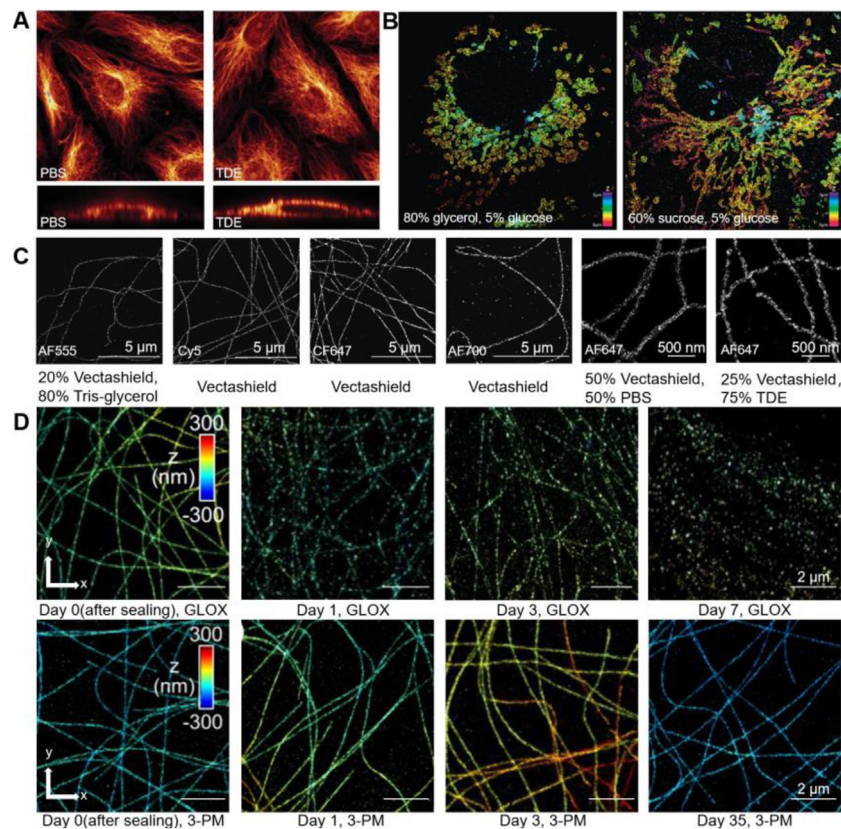


Fig. 7. Effects of refractive index regulators. (A) Confocal images of microtubules in PBS (left) and in 97% TDE (right), and corresponding x-z cross sections. Reproduced with permission [108]. Copyright 2007, Wiley-Liss. (B) Super-resolution images of mitochondria using 80% glycerol and 5% glucose (left) and 60% sucrose and 5% glucose (right). Reproduced with permission [111]. Copyright 2008, Springer Nature. (C) Super-resolution images obtained with different dyes using Vectashield as the imaging buffer. Reproduced with permission [112]. Copyright 2013, Optica Publishing Group. (D) Super-resolution images of microtubules immunostained with AF647 using different imaging buffers. Reproduced with permission [114]. Copyright 2023, American Chemical Society.

Table 1
Typical GLOX based buffers used in STORM.

Composition							Refs.
Glucose	Glucose oxidase	Catalase	Thiol/Trolox	Tris	NaCl	pH	
10% (w/v)	0.5 mg/mL	40 μg/mL	10 mmol/L (MEA)	50 mmol/L	10 mmol/L	8.0	[77]
10% (w/v)	0.5 mg/mL	40 μg/mL	35 mmol/L (MEA)	50 mmol/L	10 mmol/L	8.0	[46,48,49]
10% (w/v)	0.5 mg/mL	2000 U/mL	100 mmol/L (MEA)	50 mmol/L	10 mmol/L	8.0	[87]
10% (w/v)	0.5 mg/mL	40 μg/mL	143 mmol/L (BME)	50 mmol/L	10 mmol/L	8.0	[42,48,77]
10% (w/v)	0.5 mg/mL	40 μg/mL	1% (v/v) (BME)	50 mmol/L	10 mmol/L	7.5	[41]
0.55 mol/L	0.7 mg/mL	50 μg/mL	140 mmol/L (BME)	100 mmol/L	/	8.0	[44]
0.4% (w/w)	165 U/mL	2170 U/mL	2 mmol/L (Trolox)	10 mmol/L	50 mmol/L	8.0	[94]

Table 2
Typical imaging buffers with matched refractive index used in STORM.

Composition						Refractive index regulator	Refs.
Glucose	Glucose oxidase	Catalase	Thiol	Tris	NaCl		
8% (w/v)	0.5 mg/mL	40 μg/mL	143 mmol/L (BME)	40 mmol/L	8 mmol/L	55%–85% (v/v) (glycerol)	[48]
10% (w/v)	0.5 mg/mL	40 μg/mL	35 mmol/L (MEA)	50 mmol/L	10 mmol/L	28.5% (v/v) (TDE)	[104]
5% (w/v)	0.8 mg/mL	40 μg/mL	100 mmol/L (MEA)	50 mmol/L	3 mmol/L	82% (v/v) (3-PM)	[114]
5% (w/v)	/	/	100 mmol/L (MEA)	/	/	83% (v/v) (CUBIC-R+)	[107]

ing buffer maintained high quality for nearly five weeks, in stark contrast to samples using enzyme-based oxygen scavenging systems, such as GLOX, which only preserved quality for a few hours (Fig. 7D).

In 2024, Zhou *et al.* [107] introduced CUBIC-R+ as a high-refractive-index imaging buffer optimized for 3D dual-color STORM imaging. CUBIC-R+ is a hydrophilic chemical mixture of antipyrine with a refractive index of 1.570 and nicotinamide with a refractive

index of 1.466, which can be mixed with water to effectively match the refractive index of silicone oil or oil. Compared to 3-PM, CUBIC-R+ is colorless, less viscous, has much less absorption below the wavelength of 450 nm and much weaker emission when excited at the wavelengths of 405, 488, 561, and 642 nm. Besides, AF647 and CF680 in CUBIC-R+ have higher photon counts and photostability than those in 3-PM. The authors also demonstrated that both CUBIC-R+ and 3-PM maintained a stable pH value at 8.0 over 32 h,

distinct from the water-based buffer which exhibited a pH drop beginning at around 14 h.

Upon these discussions, we summarize typical imaging buffers with their key compositions and concentrations in Tables 1 and 2.

5. Conclusions

In this review, we present a thorough overview of the essential compositions and working mechanisms of imaging buffers in STORM. We emphasize the critical roles of oxygen scavengers, photo-switching reagents, and refractive index regulators as key components of the imaging buffer. These elements work synergistically to sustain stable blinking of fluorophores, mitigate irreversible photobleaching, and ensure high-quality super-resolution imaging results. Our review aims to serve as a valuable resource for researchers in selecting appropriate imaging buffers for their experiments.

Looking ahead, there is increasing interest in developing imaging buffers with prolonged stability to improve image resolution, reduced toxicity for compatibility with live cells, and enhanced penetrability to accommodate complex biological samples. These developments will be essential for broadening the applicability of STORM in various biological studies and further unraveling the complexities of cellular structures and dynamics within intricate tissue environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Can Wang: Writing – review & editing, Writing – original draft. **Zhe Sun:** Writing – review & editing. **Donghan Ma:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Acknowledgments

This work was funded by the National Natural Science Foundation of China (No. 62305041) and the Natural Science Foundation of Liaoning Province (No. 2023-MS-103).

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